



Modeling breakthrough curves in a fixed bed of activated carbon fiber—exact solution and parabolic approximation

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Activated carbon fiber (ACF) is a new-type high-efficiency adsorbent. It has many favourable characteristics, such as high adsorption capacity and high mass transfer rate for both adsorption and desorption, not only because of its large specific surface area and its special pore structure consisting of micropores with diameters less than 2 nm, but also because of its short diffusion path for adsorbates. In addition, ACF can be made into many forms, such as felt, sheet or honeycomb.

A number of papers have been published concerning the adsorption capacity (or adsorption equilibrium) for ACF (Kaneko *et al.*, 1988; Nitta *et al.*, 1991). Although a certain amount of information has appeared on experimental breakthrough curves for fixed-bed ACF (Yang *et al.*, 1993), theoretical research on breakthrough curves and adsorption kinetics for ACF has been scarce.

For spherical adsorbents, analytic solutions for breakthrough curves for linear adsorption systems have been derived by many authors (Rosen, 1952; Masamune and Smith, 1965; Rasmuson and Neretnieks, 1980). For simplicity, a parabolic concentration profile in the adsorbent has often been assumed (Liaw *et al.*, 1979; Rice, 1982; Cen and Yang, 1986) for fixed-bed adsorbents.

The objective of this work is to derive a general solution for breakthrough curves in a fixed bed packed with ACF and a parabolic profile approximate solution for linear adsorption systems. The effect of four independent dimensionless parameters on breakthrough curves will also be discussed.

Consider first the fixed-bed adsorber model describing an isothermal adsorption column packed with ACF. Since for ACF, the L/D ratio is large, ACF can be assumed to be cylinders of infinite length. At time zero, a step change in the concentration of an adsorbable species is introduced to the flowing stream. The adsorption column is subject to axial dispersion, mass transfer from the bulk fluid to the fiber surface, pore diffusion and adsorption at some interior site.

Material balance for the adsorbable component in the flowing phase in the fixed-bed yields the following dimensionless equation:

$$\frac{1}{Pe} \frac{\partial^2 y}{\partial \xi^2} - \frac{\partial y}{\partial \xi} - \delta \frac{\partial y}{\partial \tau} - \frac{2}{m} \delta \left(\frac{\partial y_i}{\partial \xi} \right)_{\xi=1} = 0. \quad (1)$$

The dimensionless equation for diffusion inside the ACF with one-dimensional cylindrical coordinates in the radial direction is

$$\frac{\partial^2 y_i}{\partial \xi^2} + \frac{1}{\xi} \frac{\partial y_i}{\partial \xi} = \varepsilon_p \frac{\partial y_i}{\partial \tau} + \rho_p K \frac{\partial y_q}{\partial \tau}. \quad (2)$$

For reversible adsorption, linear adsorption kinetics is assumed:

$$\frac{\partial y_q}{\partial \tau} = \psi (y_i - y_q). \quad (3)$$

The initial and boundary conditions are

$$y(0, \tau) = 1 \quad (4)$$

$$y(\xi, 0) = 0 \quad (5)$$

$$y_i(\xi, \zeta, 0) = 0 \quad (6)$$

$$y_i(0, \zeta, \tau) \neq \infty \quad (7)$$

$$y_q(\xi, \zeta, 0) = 0 \quad (8)$$

$$\left(\frac{\partial y_i}{\partial \xi} \right)_{\xi=1} = Bi[y - (y_i)_{\xi=1}]. \quad (9)$$

In this problem, the Laplace-domain solution of y can be obtained:

$$\bar{y} = \frac{1}{p} e^{r_1 \xi} \quad (10)$$

where

$$r_1 = \frac{1}{2} Pe - \sqrt{\frac{1}{4} Pe^2 + \delta p Pe + \frac{2}{m} \delta Pe H(p)} \quad (11)$$

$$H(p) = \frac{Bi \lambda I_1(\lambda)}{Bi I_0(\lambda) + \lambda I_1(\lambda)} \quad (12)$$

$$\lambda = \sqrt{\frac{\rho_p K p \psi}{p + \psi} + \varepsilon_p p}. \quad (13)$$

Following Rosen (1952), the time-domain solution is

$$y = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \exp\left(\frac{Pe}{2} \zeta - \zeta \sqrt{\frac{a^2 + b^2 + a}{2}}\right) \times \sin\left(\beta \tau - \zeta \sqrt{\frac{a^2 + b^2 + a}{2}}\right) \frac{d\beta}{\beta} \quad (14)$$

where

$$a = \frac{1}{4} Pe^2 + \frac{2}{m} \delta Pe H_1 \quad (15)$$

$$b = \delta Pe \beta + \frac{2}{m} \delta Pe H_2 \quad (16)$$

$$H_1 = \frac{Bi v^2 (1 + H_{D1}^2) + Bi^2 v (H_{D1} H_{D2} + H_{D3})}{(v H_{D1} + Bi H_{D2})^2 + (v + Bi H_{D3})^2} \quad (17)$$

$$H_2 = \frac{Bi^2 v (H_{D2} - H_{D1} H_{D3})}{(v H_{D1} + Bi H_{D2})^2 + (v + Bi H_{D3})^2} \quad (18)$$

$$H_{D1} = \frac{R_1 \cos \theta - I_{m1} \sin \theta}{R_1 \sin \theta + I_{m1} \cos \theta} \quad (19)$$

$$H_{D2} = \frac{R_0}{R_1 \sin \theta + I_{m1} \cos \theta} \quad (20)$$

$$H_{D3} = \frac{I_{m0}}{R_1 \sin \theta + I_{m1} \cos \theta} \quad (21)$$

$$R_1 = \sum_{k=0}^{\infty} \frac{1}{k!(k+1)!} \left(\frac{v}{2}\right)^{2k+1} \cos[(2k+1)\theta] \quad (22)$$

$$I_{m1} = \sum_{k=0}^{\infty} \frac{1}{k!(k+1)!} \left(\frac{v}{2}\right)^{2k+1} \sin[(2k+1)\theta] \quad (23)$$

$$R_0 = \sum_{k=0}^{\infty} \frac{1}{(k!)^2} \left(\frac{v}{2}\right)^{2k} \cos(2k\theta) \quad (24)$$

$$I_{m0} = \sum_{k=0}^{\infty} \frac{1}{(k!)^2} \left(\frac{v}{2}\right)^{2k} \sin(2k\theta) \quad (25)$$

$$v = (A_1^2 + A_2^2)^{1/4} \quad (26)$$

$$\theta = \frac{1}{2} \operatorname{atan}\left(\frac{A_2}{A_1}\right) \quad (27)$$

$$A_1 = \frac{\beta^2 \rho_p K \psi}{\psi^2 + \beta^2} \quad (28)$$

$$A_2 = \varepsilon_p \beta + \frac{\beta \rho_p K \psi^2}{\psi^2 + \beta^2} \quad (29)$$

If the adsorption rate is much faster than the diffusion rate, i.e. ψ is so large that equilibrium exists at all local positions in the fiber, adsorption equilibrium may be assumed, i.e. $y_i = y_q$. In this case, eqs (14)–(18) are still valid for the exact solution provided that eqs (19)–(29) are replaced by the following equations:

$$H_{D1} = \frac{R_1 - I_{m1}}{R_1 + I_{m1}} \quad (30)$$

$$H_{D2} = \frac{\sqrt{2}R_0}{R_1 + I_{m1}} \quad (31)$$

$$H_{D3} = \frac{\sqrt{2}I_{m0}}{R_1 + I_{m1}} \quad (32)$$

$$R_1 = \sum_{k=0}^{\infty} \frac{1}{k!(k+1)!} \left(\frac{v}{2}\right)^{2k+1} \cos\left(\frac{2k+1}{4}\pi\right) \quad (33)$$

$$I_{m1} = \sum_{k=0}^{\infty} \frac{1}{k!(k+1)!} \left(\frac{v}{2}\right)^{2k+1} \sin\left(\frac{2k+1}{4}\pi\right) \quad (34)$$

$$R_0 = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k)!^2} \left(\frac{v}{2}\right)^{4k} \quad (35)$$

$$I_{m0} = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k+1)!^2} \left(\frac{v}{2}\right)^{2(2k+1)} \quad (36)$$

$$v = \sqrt{(\rho_p K + \varepsilon_p)\beta}. \quad (37)$$

In order to reduce the complexity of the exact solution, parabolic concentration profile in the adsorbent has been assumed (Liaw *et al.*, 1979; Rice, 1982; Cen and Yang, 1986). In this work, a possible approximation for the parabolic profile is

$$y_i = a_0 + a_2 \xi^2. \quad (38)$$

Applying the boundary conditions, along with the definition of volume-averaged y_i , $(y_i)_{av}$, one obtains the dimensionless form of the parabolic approximation:

$$\frac{\partial y_i}{\partial \xi} = \frac{1}{I_1} [y - (y_i)_{av}] \quad \text{at } \xi = 1 \quad (39)$$

where

$$I_1 = \frac{1}{4} + \frac{1}{Bi}. \quad (40)$$

When the governing diffusion equation (2) is volume-averaged, one obtains

$$\frac{\partial (y_i)_{av}}{\partial \tau} = I_2 \frac{\partial y_i}{\partial \xi} \quad \text{at } \xi = 1 \quad (41)$$

where

$$I_2 = \frac{2}{\rho_p K + \varepsilon_p}. \quad (42)$$

The time-domain solution has the same form as the general solution, eqs (14)–(16). Of course, H_1 and H_2 should be replaced by the following equations:

$$H_1 = \frac{\beta^2 I_1}{\beta^2 I_1^2 + I_2^2} \quad (43)$$

$$H_2 = \frac{\beta I_2}{\beta^2 I_1^2 + I_2^2}. \quad (44)$$

The integrand of eq. (14) is the product of an exponential decaying function and a periodic sine function. Integration is performed over each half-period of the sine-wave (Rasmuson, 1985).

In the above model, eqs (1)–(9), will be tested for an actual case: $\zeta = 1$, $K\rho_p = 10,000$, $m = 10$ and $\varepsilon_p = 0.5$. The global adsorption rate is controlled by the combined effects of diffusion through the fiber void along with the local adsorption rate. The parameter ψ stands for the relative importance of the rate of adsorption and the rate of pore diffusion. The effect of ψ on the breakthrough curves is shown in Fig. 1 for $Bi = 10$, $Pe = 10$ and $\delta = 100$. The curve, calculated from eqs (14)–(18) together with eqs (30)–(37) (dashed line), is in agreement with that calculated from eqs (14)–(29) when

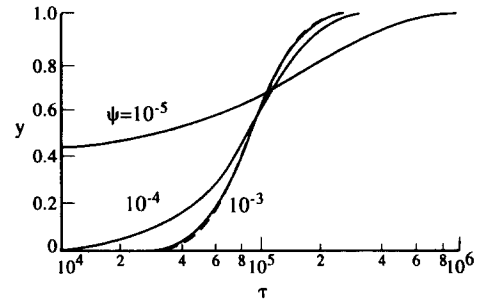


Fig. 1. Effect of parameter ψ on breakthrough curves with $Bi = 10$, $Pe = 10$ and $\delta = 100$.

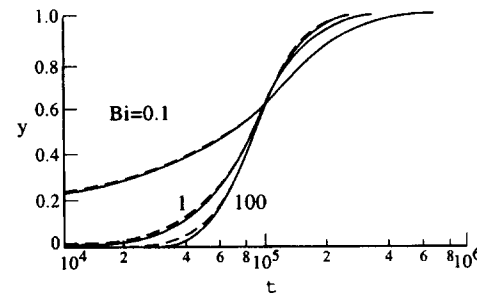


Fig. 2. Effect of parameter Bi on breakthrough curves with $Pe = 10$ and $\delta = 100$, ——— exact, - - - - parabolic.

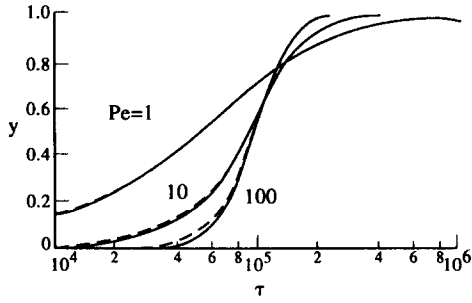


Fig. 3. Effect of parameter Pe on breakthrough curves with $Bi = 1$ and $\delta = 100$, ——— exact, - - - parabolic.

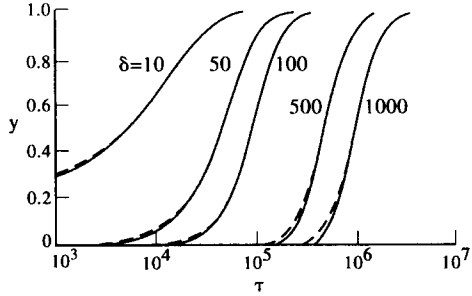


Fig. 4. Effect of parameter δ on breakthrough curves with $Bi = 1$ and $Pe = 10$, ——— exact, - - - parabolic.

$\psi = 10^{-3}$. This means that under this condition the rate of adsorption is much faster than that of pore diffusion and that equilibrium exists at every local position in the adsorbent.

Comparisons of the exact solution with the approximate solution are displayed in Figs 2, 3 and 4 for linear adsorption equilibrium. Figure 2 shows that, at $Pe = 10$ and $\delta = 100$ for three values of Bi , the parabolic approximation is in agreement with the exact solution, especially when $Bi \leq 1$. Likewise, at $Bi = 1$ and $\delta = 100$ for the three values of Pe in Fig. 3, and at $Bi = 1$ and $Pe = 10$ for the five values of δ in Fig. 4, good agreement is shown between the approximate solution and the exact solution. The agreement is also better for the smaller values of Pe or δ .

From Figs 2 and 3, one can see that if external film resistance (small Bi) or axial dispersion (small Pe) exists, the breakthrough points appear early.

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NOTATION

Bi	Biot number ($= k_f R / D_p$)
c	concentration in pore, kg/m^3
C	concentration in fluid, kg/m^3
C_0	inlet concentration in fluid, kg/m^3
D_L	longitudinal dispersion coefficient, m^2/s

D_p	pore diffusivity, m^2/s
k_a	adsorption rate constant, s^{-1}
k_f	fluid-to-fiber mass transfer coefficient, m/s
K	adsorption equilibrium constant, m^3/kg
L	height of column, m
m	$= \alpha / (1 - \alpha)$
p	Laplace transform variable
Pe	Peclet number ($= Lu / D_L$)
q	amount adsorbed, kg/kg
r	radial distance from center of fiber, m
R	fiber radius, m
t	time, s
u	flow rate, m/s
x	axial distance from column entrance, m
y	dimensionless concentration in fluid ($= C / C_0$)
$\bar{y}$	y in Laplace-domain
y_i	dimensionless concentration in pore ($= c / C_0$)
y_q	dimensionless amount adsorbed ($= q / KC_0$)

Greek letters

α	bed porosity
β	integration parameter
δ	bed length parameter ($= LD_p / uR^2$)
ϵ_p	porosity of fiber
ζ	dimensionless axial distance from column entrance ($= x / L$)
ξ	dimensionless radial distance from center of fiber ($= r / R$)
ρ_p	density of fiber, kg/m^3
τ	dimensionless time ($= D_p t / R^2$)
ψ	adsorption rate parameter ($= k_a R^2 / KD_p$)

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